

Tushar Sarap

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Summary

Embedded Software Engineer with 3+ years of experience in **Embedded C/C++**, **AUTOSAR**, and **Model-Based Development (MBD)** within automotive systems. Proven expertise in designing **real-time control software**, implementing **CI/CD pipelines (Jenkins)**, and performing **HIL/SIL validation**. Strong background in **Python-based automation**, **IBM DOORS**, and **ASPICE compliance**. Experienced in delivering scalable, safety-critical solutions across global Agile (SAFe) teams.

Technologies

Programming Languages: C++, C, Java, Python, C#

Embedded and Automotive: Embedded C/C++, AUTOSAR, CAN (Controller Area Network), JTAG Debugging, HIL (Hardware-in-the-Loop), SIL (Software-in-the-Loop), Real-Time Embedded Systems

Model-Based Development: MATLAB, Simulink, Auto-Code Generation

DevOps and Cloud: Jenkins, CI/CD, Kubernetes, Azure, Terraform

Tools and Platforms: IBM DOORS (Requirements Management), Git (Version Control), Jira, Confluence

Software Engineering Practices: Object-Oriented Programming (OOP), Agile (Scrum, SAFe), ASPICE (Automotive SPICE), V-Model, Test Automation, Integration, Debugging

Experience

Software Engineer, Bosch Global Software Technologies – Pune

Aug 2022 – Present

- Developed **embedded software in C/C++** for automotive ECUs aligned with **AUTOSAR architecture and ASPICE standards**.
- Designed **model-based solutions using MATLAB/Simulink**, enabling **auto-code generation** and improving development efficiency.
- Built and optimized **CI/CD pipelines using Jenkins**, reducing deployment time by **30%** and improving release stability.
- Led **HIL testing** for safety-critical modules, ensuring compliance with **functional safety and reliability requirements**.
- Developed **Python-based automation frameworks**, improving regression coverage and reducing manual QA effort.
- Performed **software integration, debugging, and root cause analysis** across application, RTE, and BSW layers.
- Maintained **100% requirement traceability** for 200+ requirements using **IBM DOORS**, ensuring audit readiness.
- Collaborated in **Agile (SAFe)** teams for feature development, integration, and global release delivery.

Projects

Model-Based Development of Brake Control Function

- Developed a **brake control function using MATLAB/Simulink (MBD)** for automotive applications.
- Implemented **Stateflow-based state machines** and control logic for braking and safety mechanisms.
- Performed **MIL and SIL validation** to verify functional correctness and system behavior.
- Generated **embedded C code using Simulink Coder** and integrated into ECU software.
- Validated functionality using **HIL testing**, ensuring real-time system performance.

Embedded ECU Development and Automated Validation Framework

- Developed **embedded C/C++ software components** following **AUTOSAR and MISRA guidelines**.
- Implemented **application logic, state machines, and CAN communication interfaces**.
- Built **Python-based validation frameworks** with reusable modules for automated ECU testing.
- Integrated testing into **CI/CD pipelines (Jenkins)** for continuous validation.

- Improved **test coverage and early defect detection** across development cycles.

CI/CD Platform for Embedded Software Delivery

- Designed **CI/CD pipelines (Jenkinsfiles)** for multi-variant embedded builds.
- Developed **automation scripts (Python/Shell)** for build orchestration and deployment.
- Integrated **HIL/SIL testing into pipelines** for continuous validation.
- Leveraged **Kubernetes, Azure, and Terraform** for scalable DevOps workflows.
- Reduced **integration time** and improved release frequency.

Brake System Software Development (Toyota Variant Learning)

- Developed **safety-critical braking software** using Embedded C/C++.
- Implemented **control logic and feature enhancements** for multiple vehicle variants.
- Performed **low-level debugging (JTAG)** and resolved critical real-time issues.
- Increased **code coverage** and validation aligned with **ISO 26262 practices**.
- Improved **scalability and reuse** across variants, reducing re-engineering effort.

Education

SGGS College of Engineering & Technology, Nanded

Aug 2018 – May 2022

- **Course:** Bachelor of Technology in Electrical Engineering